

Software Test Plan (STP)

Hospital Management System (HMS) – Patient Record & Appointment Management

Field	Details
Project	Hospital Management System (HMS)
Team	Team 13
Version	1.0
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Status	Draft
Based on	HMS SRS v1.0 (06-09-2026);

Note on SRS requirement count

The SRS note states 22 Functional Requirements, but the requirement tables list 23 (F-001 to 003, 010 to 014, 020 to 024, 030 to 032, 040 to 041, 050 to 051, 060 to 062). Correct this in the SRS so the RTM and STP counts agree. This plan covers 23 FRs, 6 NFRs and 5 security requirements.

1. Introduction

- **Purpose:** Test plan for HMS v1.0 (C/C++, SQLite, optional TCP LAN mode). It defines objectives, scope, strategy, resources, schedule and responsibilities for testing.
- **Scope:** All 23 functional requirements, 6 non-functional requirements and 5 security requirements in the HMS SRS.
- **References:** HMS SRS v1.0 (06-09-2026), the design document, and the course guidelines.
- **Definitions:** The SRS definitions apply (HMS, OPD, UI, CLI, DB, RBAC, RTM, FR, NFR), plus STP (Software Test Plan), TC (Test Case), UAT (User Acceptance Testing) and DAL (Data Access Layer).

2. Test Items

The modules under test match the design components in the SRS RTM:

- Auth Module
- Patient Manager
- Appointment Manager
- Doctor Manager
- Consultation Manager
- Billing Manager

- Audit Logger
- Reporting/Admin (backup, health check)
- Data Access Layer (SQLite)
- TCP Server/Client

3. Features to be Tested

Every requirement ID in the SRS is tested. The full mapping of requirement to priority and test case is in Section 13.

4. Features Not to be Tested

- Optional Qt/GTK GUI (stretch goal)
- Barcode/ID-card reader (future enhancement)
- Printer hardware (only the slip/bill content is tested)
- SQLite engine internals
- OS and socket libraries
- Out-of-scope ERP functions (pharmacy, insurance, lab)

5. Test Approach / Strategy

Test levels

- **Unit:** One test file per module, using Unity, CUnit, or GoogleTest/Catch2 for C++.
- **Integration:** UI to business logic to DAL to SQLite, and client to server over TCP.
- **System:** End-to-end role workflows, e.g. Receptionist registers a patient and books, Doctor prescribes, then billing.
- **Acceptance:** Run against the criteria in SRS Section 6.

Test types

Functional, regression, performance (NF-001), usability (NF-003), reliability/crash (NF-004), portability (NF-005), and concurrency (two clients booking the same slot).

Entry criteria

A stable build that compiles on Windows and Linux, and a seeded test database ready.

Exit criteria

Matches SRS Section 6: all High-priority FRs pass, 0 critical defects open, all 5 security requirements verified, NF-001 and NF-004 demonstrated, and builds pass on both platforms.

5.1 Security Validation

- Inspect the DB for plaintext passwords (SR-001).
- Call DAL functions directly with the wrong role (SR-002).
- Check that the audit log has an entry for every create/update/delete (SR-003).
- Run fuzz/long-string tests on every input field, plus SQL injection strings such as ' OR 1=1 -- (SR-004).
- Run static analysis (cppcheck) and sanitizers (ASan/Valgrind).
- Check the 10-minute idle logout (SR-005).
- Test unauthenticated TCP requests; the server must reject them.

6. Test Environment

- **Hardware:** One Windows machine and one Linux machine (or VMs), plus 2 or more clients for LAN mode.
- **Software:** HMS build, SQLite3, and a seed script for 10,000 patients.
- **Tools:** Unity/CUnit/GoogleTest; gcov for coverage; Valgrind or ASan for memory bugs; cppcheck; AFL++ or libFuzzer for fuzzing; a Python socket script for protocol tests; GitHub Issues in place of Jira.
- **Test data:** Dummy patients, doctors and staff in every role. Never use real patient data.

7. Test Schedule

Milestone	Date
Test case design	05-10-2026 to 12-10-2026
Environment setup	13-10-2026 to 16-10-2026
Test execution start	19-10-2026
Test execution end	30-10-2026
UAT	02-11-2026 to 05-11-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated)
- Test Scripts
- Test Data, including the seed-data script
- Test Execution Logs
- Defect Reports
- Coverage (gcov) and static-analysis reports
- Test Summary Report

9. Roles and Responsibilities

Fill in the team member name against each role. With a small team, one person can hold more than one role.

Role	Name	Responsibility
Tester	B M Harshith Kumar	Design and execute test cases, run regression and security/fuzz tests, log defects, prepare the Test Summary Report
Developer	Vipin V Lokesh	Write unit tests, fix defects, support triage, keep the build compiling on Windows and Linux
Documentation/SRS Lead	Madhu Sripada	Write and maintain the SRS (including correcting the FR count) and keep the RTM and this STP in sync with it
Database/Backend Lead	Chandana O N	Own the SQLite schema, seed script (10,000 patients), data access layer, TCP server and backup/restore support for test environments

10. Risks and Mitigation

Risk	Mitigation
Time-based tests (5-min lockout, 10-min idle) are slow	Make the timeouts configurable, or inject a mock clock in tests
Race condition on double-booking in LAN mode	Dedicated two-client concurrency test; use DB transactions/unique constraint
Windows/Linux build differences	Run CI or a manual build on both early
Small team, limited time	Prioritize High-priority requirements first
Confidentiality over LAN: the SRS has no transport-encryption requirement	Note it as a limitation or add an SR

11. Assumptions & Dependencies

- The DB schema is final before execution.
- Test data is seeded before execution.
- Both Windows and Linux environments are available.

12. Suspension & Resumption Criteria

- **Suspend if:** the build fails to compile, the DB is corrupted, or more than 30% of test cases are blocked.
- **Resume when:** these issues are fixed and the environment is stable.

13. Test Case Management & Traceability

This RTM expands the SRS RTM (which covers only 8 requirements) to every requirement.

Req ID	Requirement	Priority	Test Case
HMS-F-001	Staff authentication (username/password)	High	TC-Auth-01
HMS-F-002	Role-based access control	High	TC-Auth-02
HMS-F-003	Account lockout after 3 failed logins	High	TC-Auth-03
HMS-F-010	Register new patient	High	TC-Pat-01
HMS-F-011	Search patient by ID/name/phone	High	TC-Pat-02
HMS-F-012	Update patient details	Medium	TC-Pat-03
HMS-F-013	View patient visit and prescription history	High	TC-Pat-04
HMS-F-014	Soft-delete (archive) patient record	Medium	TC-Pat-05
HMS-F-020	Book appointment	High	TC-Appt-01
HMS-F-021	Doctor availability check / reject conflicts	High	TC-Appt-02
HMS-F-022	Reschedule or cancel appointment	High	TC-Appt-03
HMS-F-023	Doctor daily/weekly schedule	High	TC-Appt-04
HMS-F-024	Appointment confirmation slip	Medium	TC-Appt-05
HMS-F-030	Manage doctor profiles	High	TC-Doc-01
HMS-F-031	Create staff accounts with role	High	TC-Doc-02
HMS-F-032	Doctor patient queue/workload	Medium	TC-Doc-03
HMS-F-040	Record diagnosis and prescription	High	TC-Rx-01
HMS-F-041	Prescription history by visit date	Medium	TC-Rx-02
HMS-F-050	Generate consultation bill	High	TC-Bill-01

Req ID	Requirement	Priority	Test Case
HMS-F-051	Payment status (Paid/Pending)	Medium	TC-Bill-02
HMS-F-060	Daily appointment report	Medium	TC-Rep-01
HMS-F-061	Database backup/export	High	TC-Rep-02
HMS-F-062	Admin health check	Medium	TC-Rep-03
HMS-NF-001	Lookup under 2 s for 10,000 records	High	TC-Perf-01
HMS-NF-002	Availability (max 1 unplanned restart/week)	High	TC-Avail-01
HMS-NF-003	Usable with under 30 min training	Medium	TC-Use-01
HMS-NF-004	No data loss on crash/power loss	High	TC-Rel-01
HMS-NF-005	Builds and runs on Windows and Linux	Medium	TC-Port-01
HMS-NF-006	Modular code (UI, logic, DAL)	Medium	TC-Maint-01
HMS-SR-001	Salted password hashes	High	TC-Sec-01
HMS-SR-002	RBAC enforced at data-access layer	High	TC-Sec-02
HMS-SR-003	Audit log of create/update/delete	High	TC-Sec-03
HMS-SR-004	Input validation / bound checking	High	TC-Sec-04
HMS-SR-005	Auto logout after 10 min idle	Medium	TC-Sec-05

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage
- Code coverage (gcov)

Reports: daily execution status and the final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
Tester	B M Harshith Kumar	
Developer	Madhu Shripada	
Documentation/SRS Lead	Vipin V Lokesh	
Database/Backend Lead	Chandana O N	
Course Coordinator	Proff.Rajeshwari Adarkatti	

Appendix A: Detailed Test Cases (examples)

The STP lists the TC IDs; this appendix shows the level of detail expected for each test case.

TC ID	Req	Steps	Expected result
TC-Auth-03	F-003	Enter a wrong password 3 times, then try the correct one immediately	Lockout message shown; audit event logged; login works after 5 minutes

TC ID	Req	Steps	Expected result
TC-Appt-02	F-021	Book Dr. A at 10:00 on a date; book Dr. A at 10:00 again	Second booking rejected with a message
TC-Appt-02b	F-021	Two LAN clients book the same slot at the same moment	Exactly one booking succeeds
TC-Perf-01	NF-001	Seed 10,000 patients; search by ID, name and phone 50 times	Every lookup completes in under 2 s
TC-Rel-01	NF-004	Kill the process during a patient insert, then restart	DB intact, no partial record
TC-Sec-04	SR-004	Enter 5,000 characters and SQL strings into every input field	No crash, no overflow, no SQL execution

Negative and boundary cases to add

- Empty fields
- Invalid phone number
- Age 0 or 150
- The exact 3rd and 4th login attempts
- The 9:59 vs 10:00 idle-timeout boundary